Turning Probability into Assets: A Look Ahead at Prediction Market Agents

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In our previous Crypto AI research, we established that while **stablecoins** and **DeFi** offer immediate utility, Agents represent the critical user interface for the AI industry. Consequently, we define two primary value paths for Crypto-AI integration: a short-term focus on **AgentFi**, which automates yield strategies on mature DeFi protocols, and a medium-to-long-term evolution toward **Agent Payment**, enabling autonomous stablecoin settlement via emerging standards like ACP, x402, and ERC-8004.

Prediction markets have become an undeniable new industry trend in 2025, with total annual trading volume surging from approximately $9 billion in 2024 to over $40 billion in 2025, achieving a year-on-year growth of over 400%. This significant growth is driven by multiple factors: demand for uncertainty hedging brought by macro-political events, the maturation of infrastructure and trading models, and the breaking of ice in the regulatory environment (Kalshi's lawsuit victory and Polymarket's return to the US). **Prediction Market Agents** are showing early prototypes in early 2026 and are poised to become a new product form in the agent field over the coming year.

I. Prediction Markets: From Betting Tools to a "Global Truth Layer"

A prediction market is a financial mechanism for trading around the outcomes of **future events**. Contract prices essentially reflect the market's collective judgment on the probability of an event occurring. Its effectiveness stems from the combination of **crowd wisdom** and **economic incentives**: in an environment of anonymous, real-money betting, dispersed information is rapidly integrated into price signals weighted by financial willingness, thereby significantly reducing noise and false judgments.

(Note: "Prediction Market Nominal Trading Volume Trend Chart" from Dune Analytics here.)

By the end of 2025, prediction markets have largely formed a duopoly dominated by **Polymarket** and **Kalshi**. According to *Forbes*, total trading volume in 2025 reached approximately $44 billion, with Polymarket contributing about $21.5 billion and Kalshi about $17.1 billion. February 2026 weekly data shows Kalshi's trading volume ($25.9B) has surpassed Polymarket ($18.3B), approaching 50% market share. Kalshi, leveraging its legal victory in the previous election contract case, its first-mover compliance advantage in the US sports prediction market, and relatively clear regulatory expectations, achieved rapid expansion. Currently, their development paths have clearly diverged:

> **@Polymarket** adopts a hybrid CLOB (Central Limit Order Book) architecture with "off-chain matching, on-chain settlement" and a decentralized settlement mechanism. It has built a globalized, non-custodial high-liquidity market, forming an "onshore + offshore" dual-track operational structure after its compliant return to the US.
>
> **@Kalshi** integrates into the traditional financial system, accessing mainstream retail brokers via API to attract Wall Street market makers for deep participation in macro and data-based contract trading. Its products are constrained by traditional regulatory processes, leading to a lag in addressing long-tail demands and sudden events.

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Beyond Polymarket and Kalshi, other competitive participants in the prediction market field are developing along two main paths:

Compliant Distribution Path: Embedding event contracts into the existing account and clearing systems of brokers or large platforms, relying on channel coverage, compliance qualifications, and institutional trust to build advantages (e.g., Interactive Brokers × ForecastEx's ForecastTrader, FanDuel × CME Group's FanDuel Predicts). While compliance and resource advantages are significant, product and user scale are still in the early stages.

Crypto-Native On-Chain Path: Represented by @opinionlabsxyz, @trylimitless and @MyriadMarkets, these leverage points mining, short-cycle contracts, and media distribution to achieve rapid volume growth. They emphasize performance and capital efficiency, but their long-term sustainability and risk control robustness remain to be verified.

These two paths—traditional financial compliance entry and crypto-native performance advantages—together constitute the diversified competitive landscape of the prediction market ecosystem.

While prediction markets superficially resemble gambling and are essentially zero-sum games, the core difference lies in whether they possess **positive externalities**: aggregating dispersed information through real-money trading to publicly price real-world events, forming a valuable signal layer. The trend is shifting from gaming to a "**Global Truth Layer**"—as institutions like CME and Bloomberg connect, event probabilities have become decision-making metadata directly callable by financial and corporate systems, providing a more timely, quantifiable, market-based truth.

From a global regulatory perspective, compliance paths for prediction markets are highly divergent. The US is the only major economy explicitly including prediction markets in its financial derivatives regulatory framework. Markets in Europe, the UK, Australia, and Singapore generally view them as gambling and tend to tighten regulations, while China and India completely ban them. Future global expansion of prediction markets still depends on national regulatory frameworks.

II. Architecture Design of Prediction Market Agents

Prediction Market Agents are currently entering an early practice stage. Their value lies not in "AI predicting more accurately," but in amplifying information processing and execution efficiency within prediction markets. Prediction markets are essentially information aggregation mechanisms

where price reflects the collective judgment of event probability; real-world market inefficiencies stem from information asymmetry, liquidity, and attention constraints. The reasonable positioning for a Prediction Market Agent is **Executable Probabilistic Portfolio Management**: converting news, rule texts, and on-chain data into verifiable pricing deviations, executing strategies in a faster, more disciplined, and lower-cost manner, and capturing structural opportunities through cross-platform arbitrage and portfolio risk control.

An ideal Prediction Market Agent can be abstracted into a four-layer architecture:

> **Information Layer:** Aggregates news, social media, on-chain, and official data.
>
> **Analysis Layer:** Uses LLMs and ML to identify mispricing and calculate Edge.
>
> **Strategy Layer:** Converts Edge into positions using the Kelly Criterion, staggered entry, and risk control.
>
> **Execution Layer:** Completes multi-market order placement, slippage and Gas optimization, and arbitrage execution, forming an efficient automated closed loop.

III. Strategy Framework for Prediction Market Agents

Unlike traditional trading environments, prediction markets have significant differences in **settlement mechanisms**, **liquidity**, and **information distribution**. Not all markets and strategies are suitable for automated execution. The core of a Prediction Market Agent lies in whether it is deployed in scenarios with clear rules, codifiability, and structural advantages. The following analysis covers target selection, position management, and strategy structure.

1. Prediction Market Target Selection

Not all prediction markets have tradable value. Participation value depends on: **Settlement Clarity (are rules clear, is the data source unique)**, **Liquidity Quality (market depth, spread, and volume)**, **Insider Risk (degree of information asymmetry)**, Time Structure (expiration time and event pacing), and the trader's own Information Advantage and Professional Background. A prediction market only has a basis for participation when most dimensions meet basic requirements. Participants should match based on their own strengths and market characteristics:

Human Core Advantage: Markets relying on domain expertise, judgment, and integration of ambiguous information, with relatively loose time windows (days/weeks). Typical examples: Political elections, macro trends, and corporate milestones.

AI Agent Core Advantage: Markets relying on data processing, pattern recognition, and rapid execution, with extremely short decision windows (seconds/minutes). Typical examples: High-frequency crypto prices, cross-market arbitrage, and automated market making.

Unsuitable Areas: Markets dominated by insider information or purely random/highly manipulated markets, which offer no advantage to any participant.

2. Position Management in Prediction Markets

The **Kelly Criterion** is the most representative capital management theory in repeated games. Its goal is not to maximize the return of a single trade, but to maximize the long-term compound growth rate of capital. It calculates the theoretical optimal position ratio based on estimates of win rate and odds, improving capital growth efficiency under the premise of positive expectancy. It is widely used in quantitative investment, professional gambling, poker, and asset management.

> **Classic Formula:** $f^* = (bp - q) / b$
>
> Where $f*$ is optimal betting fraction, b is net odds, p is win rate, and $q=1-p$.
>
> **Simplified for PM:** $f^* = (p - market_price) / (1 - market_price)$
>
> Where p is the subjective true probability, market_price is the market implied probability.

The theoretical effectiveness of the Kelly formula is highly dependent on accurate estimates of true probability and odds. In reality, traders find it difficult to consistently and accurately grasp the true probability. In practice, professional gamblers and prediction market participants tend to adopt rule-based strategies that are more executable and less dependent on probability estimation:

> **Unit System:** Splits capital into fixed units (e.g., 1%) and invests different numbers of units based on confidence levels. This automatically constrains single-bet risk through a unit cap and is the most common practical method.
>
> **Flat Betting:** Uses a fixed percentage of capital for each bet. Emphasizes discipline and stability, suitable for risk-averse or low-conviction environments.
>
> **Confidence Tiers:** Presets discrete position tiers and sets absolute caps to reduce decision complexity and avoid the false precision problem of the Kelly model.
>
> **Inverted Risk Approach:** Calculates position size backwards starting from the maximum tolerable loss. It defines boundaries from risk constraints rather than profit expectations.

For Prediction Market Agents, strategy design should prioritize **executability and stability** over theoretical optimality. The key lies in clear rules, simple parameters, and tolerance for judgment errors. Under these constraints, the **Confidence Tiers method combined with fixed position caps** is the most suitable general position management scheme for PM Agents. This method does not rely on precise probability estimates but divides opportunities into limited tiers based on signal strength, setting clear caps to control risk even in high-conviction scenarios.

3. Strategy Selection for Prediction Markets

Structurally, strategies fall into two main categories: **Deterministic Arbitrage** strategies (characterized by clear rules and codifiability) and **Speculative Directional** strategies (relying on information interpretation and direction judgment). Additionally, there are Market Making and Hedging strategies, mainly for professional institutions with high capital and infrastructure requirements.

Deterministic Arbitrage Strategies (Arbitrage)

Resolution Arbitrage: Occurs when an event outcome is basically determined but the market hasn't fully priced it in yet. Returns come from information synchronization and execution speed. Rules are clear, risk is low, and it is fully codifiable—the core strategy most suitable for Agent execution.

Dutch Book Arbitrage (Probability Conservation): Exploits structural imbalances where the sum of prices for a mutually exclusive and exhaustive set of events deviates from the probability conservation constraint ($\sum P \neq 1$). By building a portfolio, it locks in risk-free returns. It relies only on rules and price relationships, has low risk, and can be highly regularized. It is a typical deterministic arbitrage form suitable for automated Agent execution.

Cross-Platform Arbitrage: Profits by capturing pricing deviations for the same event across different markets. Low risk but high requirements for latency and parallel monitoring. Suitable for Agents with infrastructure

advantages, but competition is intensifying, leading to declining marginal returns.

Bundle Arbitrage: Exploits pricing inconsistencies between related contracts. Logic is clear but opportunities are limited. Can be executed by Agents but requires some engineering for rule parsing and portfolio constraints. Agent suitability is medium.

Speculative Directional Strategies (Speculative)

Structured Information Driven (Information Trading): Centers around clear events or structured information, such as official data releases, announcements, or ruling windows. As long as the information source is clear and trigger conditions are definable, Agents can leverage speed and discipline in monitoring and execution. However, when information turns into semantic judgment or scenario interpretation, human intervention is still needed.

Signal Following: Profits by following accounts or capital behaviors with historically superior performance. Rules are relatively simple and automatable. The core risk lies in signal decay and being front-run/counter-traded, requiring filtering mechanisms and strict position management. Suitable as an auxiliary strategy for Agents.

Unstructured / Noise-driven: Highly dependent on sentiment, randomness, or participation behavior. Lacks a stable, reproducible edge, and long-term expected value is unstable. Difficult to model and extremely high risk; not suitable for systematic Agent execution and not recommended as a long-term strategy.

High-Frequency Price & Liquidity Strategies (Market Microstructure): Relies on extremely short decision windows, continuous quoting, or high-frequency trading. Requirements for latency, models, and capital are extremely high. While theoretically suitable for Agents, they are often limited by liquidity and competition intensity in prediction markets, suitable only for a few participants with significant infrastructure advantages.

Risk Control & Hedging: Does not directly seek profit but is used to reduce overall risk exposure. Clear rules and objectives; runs long-term as an underlying risk control module.

Summary: Strategies suitable for Agent execution in prediction markets are concentrated in scenarios with **clear rules, codifiability, and weak subjective judgment**. Deterministic arbitrage should be the core revenue source, with structured information and signal following strategies as supplements. High-noise and emotional trading should be systematically excluded. An Agent's long-term advantage lies in disciplined, high-speed execution and risk control capabilities.

IV. Business Models and Product Forms of Prediction Market Agents

Ideal business model designs for Prediction Market Agents have exploration space at different levels:

Infrastructure Layer: Provides multi-source real-time data aggregation, Smart Money address libraries, unified prediction market execution engines, and backtesting tools. Charges B2B fees to obtain stable revenue unrelated to prediction accuracy.

Strategy Layer: Introduces community and third-party strategies to build a reusable, evaluable strategy ecosystem. Captures value through calls, weights, or execution profit-sharing, reducing dependence on a single Alpha.

Agent / Vault Layer: Agents directly participate in live trading via entrusted management, relying on on-chain transparent records and strict risk control systems to earn management fees and performance fees based on capability.

Corresponding product forms can be divided into:

Entertainment / Gamification Mode: Lowers participation barriers through Tinder-like intuitive interaction. Has the strongest user growth and market education capability, making it an ideal entry point for breaking out of the niche, but needs to funnel users to subscription or execution products for monetization.

Strategy Subscription / Signal Mode: Does not involve capital custody, is regulatory-friendly with clear rights and responsibilities, and has a relatively stable SaaS revenue structure. It is currently the most feasible commercialization path. Its limitation is that strategies are easily copied and execution suffers from slippage. Long-term revenue ceilings are limited, but experience and retention can be significantly improved through a "Signal + One-Click Execution" semi-automated form.

Vault Custody Mode: Possesses scale effects and execution efficiency advantages, resembling asset management products. However, it faces multiple structural constraints such as asset management licenses, trust thresholds, and centralized technical risks. The business model is highly dependent on the market environment and sustained profitability. Unless possessing a long-term track record and institutional-grade endorsement, it should not be the main path.

Overall, a diversified revenue structure of **"Infrastructure Monetization + Strategy Ecosystem Expansion + Performance Participation"** helps reduce reliance on the single assumption that "AI consistently beats the market." Even if Alpha converges as the market matures, underlying capabilities like execution, risk control, and settlement retain long-term value, thus building a more sustainable business closed loop.

V. Project Cases of Prediction Market Agents

Currently, Prediction Market Agents are still in the early exploration stage. Although the market has seen diverse attempts from underlying frameworks to upper-layer tools, a standardized product that is mature in strategy generation, execution efficiency, risk control systems, and business closed loops has not yet formed.

We classify the current ecosystem landscape into three levels: **Infrastructure**, **Autonomous Agents**, and **Prediction Market Tools**.

Infrastructure Layer

Polymarket Agents Framework @Polymarket

This official developer framework standardizes "connection and interaction," handling data retrieval, order construction, and basic LLM interfaces. However, it functions primarily as an access standard rather than a turnkey solution; it solves "how to code an order" but leaves core trading capabilities—such as strategy generation, probability calibration, and risk management—entirely to the developer.

Gnosis Prediction Market Tools

@gnosis_ Offering complete read/write support for the Gnosis ecosystem (Omen/Manifold), this toolset provides only read access for Polymarket,

creating clear ecosystem barriers. It serves as a strong foundation for Gnosis-native agents but has limited utility for cross-platform development.

Polymarket and Gnosis are currently the only prediction market ecosystems that have clearly productized "Agent Development" into official frameworks. Other prediction markets like Kalshi still mainly remain at the API and Python SDK level, requiring developers to self-complete key system capabilities like strategy, risk control, operation, and monitoring.

Autonomous Agents

Current "Prediction Market AI Agents" on the market are mostly still in early stages. Although labeled "Agent," their actual capabilities are significantly far from delegatable automated closed-loop trading. They generally lack independent, systematic risk control layers and have not incorporated position management, stop-loss, hedging, and expected value constraints into the decision process. Overall productization is low, and mature systems for long-term operation have not yet formed.

Olas Predict @autonolas

Olas Predict is currently one of the most productized prediction market agent ecosystems. Its core product **Omenstrat**, built on Omen within the Gnosis ecosystem, utilizes FPMM and decentralized arbitration mechanisms, enabling small-scale high-frequency interactions, though overall scale is constrained by Omen's relatively limited single-market liquidity. Its AI-driven predictions rely on LLM-based tooling combined with available online information sources to derive probability estimates, while overall performance has generally trended positive for many users.

Polystrat, built on Olas, launched in February 2026 and extends agent capabilities to Polymarket. Users can define strategies in natural language, and the agent identifies probability deviations in markets settling within four days and executes trades automatically. The system manages risk through local execution via **@pearldotyou** (available on Mac, Windows, and Linux), self-custodied **@Safe smart accounts**, and hardcoded safety limits—making it one of the first consumer-grade autonomous trading agents available for Polymarket.

UnifAI Network Polymarket Strategy @UnifaiNetwork

Provides automated trading Agent for Polymarket, with a core **tail risk strategy**: scanning contracts near settlement with >95% implied probability and buying in, targeting 3–5% spread capture. On-chain data shows a win rate close to 95%, but returns diverge significantly across categories. The strategy is highly dependent on execution frequency and category selection.

NOYA.ai @NetworkNoya

Attempts a comprehensive "Research-Judgment-Execution" closed loop. Its architecture features an Intelligence Layer for signal aggregation and an Abstraction Layer using Intents to manage cross-chain complexity. Currently,

its Omnichain Vaults have been delivered; the Prediction Market Agent remains under development, and a complete mainnet closed loop has not yet formed. Overall, it is in the vision validation stage.

Prediction Market Tools

Current prediction market analysis tools are insufficient to constitute complete "Prediction Market Agents." Their value is mainly concentrated in the Information and Analysis layers of the agent architecture; trade execution, position management, and risk control must still be borne by the trader. Product forms align more with "Strategy Subscription / Signal Assistance / Research Enhancement" and can be viewed as early prototypes of Prediction Market Agents.

Based on a systematic review of Awesome-Prediction-Market-Tools, we selected representative projects with preliminary product forms:

Market Analysis Tools

Polyseer : Research-oriented tool using a multi-Agent architecture (Planner/Researcher/Critic/Analyst/Reporter) for evidence collection and Bayesian aggregation to output structured reports. Transparent methodology, open-source.

Oddpool: "Bloomberg Terminal for Prediction Markets," aggregating Polymarket, Kalshi, CME, etc., with arbitrage scanning.

Polymarket Analytics: Global data analysis platform for Polymarket, showing trader, market, position, and volume data.

Hashdive @hash_dive **:** Trader-oriented data tool using Smart Score to identify "Smart Money."

Polyfactual @polyfactual **:** Focuses on AI market intelligence and sentiment/risk analysis via Chrome extension.

Predly: AI mispricing detection platform comparing market prices with AI-calculated probabilities on Polymarket and Kalshi. Claims 89% alert accuracy.

Polysights: Covers 30+ markets and on-chain metrics with Insider Finder tracking new wallets and large unidirectional bets.

PolyRadar: Multi-model parallel analysis with real-time interpretation, timeline evolution, and confidence scoring.

Alphascope: AI-driven intelligence engine for real-time signals and research summaries (early stage).

Alerts / Whale Tracking

Stand: Focuses on whale copy-trading and high-conviction alerts.

Whale Tracker Livid : Productizes whale position changes.

Arbitrage Discovery Tools

ArbBets @arbbets **:** AI-driven tool identifying cross-platform arbitrage (Polymarket, Kalshi, Sportsbooks).

PolyScalping @PolyScalping**:** Real-time arbitrage and scalping analysis for Polymarket (1-minute scans).

Eventarb @eventarbitrage**:** Lightweight cross-platform arbitrage calculator (Polymarket, Kalshi, Robinhood).

Prediction Hunt: Cross-exchange aggregator comparing prices for arbitrage (Polymarket, Kalshi, PredictIt).

Trading Terminals / Aggregated Execution

Verso: Institutional-grade terminal (YC Fall 2024) with Bloomberg-style interface, covering 15,000+ contracts across Polymarket and Kalshi with AI news intelligence.

Matchr @matchrxyz: Cross-platform aggregator covering 1,500+ markets with smart routing for optimal price matching and planned automated yield strategies.

TradeFox: Professional aggregation and Prime Brokerage platform backed by Alliance DAO and CMT Digital. Offers advanced order execution (limit, stop-loss, TWAP), self-custody, and multi-platform smart routing. Expanding to Kalshi, Limitless, and SxBet.

VI. Summary and Outlook

Currently, Prediction Market Agents are in the early exploration stage of development.

Market Essence: Backed by the Polymarket and Kalshi duopoly, prediction markets differ from gambling by acting as a "**Global Truth Layer**" that aggregates information via real-money trading.

Core Positioning: Agents function as **Executable Probabilistic Portfolio Management** tools. They convert data into verifiable pricing deviations, prioritizing discipline and execution speed.

Strategy & Risk: Deterministic Arbitrage is the optimal strategy for automation, with speculation serving only as a supplement. Risk management should prioritize executability using **Confidence Tiers with Fixed Caps**.

Business Model: The most sustainable path combines **Infrastructure** (Data/execution fees), **Strategy Ecosystems** (third-party licensing), and **Vaults** (performance-based asset management).

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Despite the emergence of diverse tools and frameworks in the ecosystem, a mature, standardized product capable of closing the loop on strategy generation, execution efficiency, and risk control has yet to appear. We look forward to the continued iteration and evolution of Prediction Market Agents.

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